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1. **Purpose**

   This matrix links every use requirement (URS) to its functional requirements (FRS), its risk level from the FMEA (part 2), and the test that proves it (IQ, OQ, PQ, or a named report). A requirement with no test is a gap. IQ and PQ are not yet written, so their entries show the planned phase only.

2. **Requirements to tests**

URS ID	FRS ID	Risk Level	Test(s)
1.1	N/A	Low	IQ Document review
1.2	1.2.1, 1.2.2	Low	IQ Datasheet and room record
1.3	1.3.1	Low	IQ Measurement
1.4	N/A	Low	IQ Measurement
1.5	N/A	Low	IQ Weight certificate
1.6	N/A	Low	IQ Calculation
2.1	2.1.1,2.1.2,2.1.3,2.1.4,	High	OQ-4
2.2	2.2.1,2.2.2,2.2.3,2.2,4,2.2.5	High	OQ-15
2.3	2.3.1,2.3.2,2.3.3	Medium	OQ-14
2.4	2.4.1,2.4.2,2.4.3	Medium	OQ-14
2.5	2.5.1	High	OQ-10
3.1	3.1.1,3.1.2,3.1.3	High	OQ-01
3.2	3.2.1	Low	OQ-02
3.3	3.3.1,3.3.2	Medium	OQ-02
3.4`	3.4.1	Low	OQ-02
3.5	3.5.1	Low	Unscripted OQ check B
4.1	4.1.1,4.1.2,4.1.3,4.1.4	High	IQ certificate and OQ-5
4.2	4.2.1,4.2.2,4.2.3	Medium	OQ-07
4.3	4.3.1,4.3.2,4.3.3	High	OQ-15
4.4	4.4.1,4.4.2	Medium	OQ-04
5.1	5.1.1,5.1.2	Medium	OQ-16
5.2	5.2.1,5.2.2	Low	OQ
6.1	6.1.1,6.1.2,6.1.3	High	OQ-06 and PQ
6.2	6.2.1,6.2.2	High	Steam quality test
6.3	6.3.1	Low	IQ and PQ
6.4	6.4.1,6.4.2	Low	OQ-04
7.1	7.1.1,7.1.2	Medium	OQ-12
7.2	7.2.1	High	OQ-12
7.3	7.3.1,7.3.2	High	OQ-11
8.1	8.1.1	Medium	OQ-17

8.2	8.2.1	High	OQ-03
8.3	8.3.1,8.3.2	High	OQ-03
8.4	8.4.1	Medium	OQ-16
9.1	9.1.1	Low (severity override)	OQ-13
9.2	9.2.1	Low	OQ-09
9.3	9.3.1	Low	REVIEW OPERATIONS
9.4	9.4.1	Low	IQ document review

3. Test to requirements (reverse view)

Test	Requirements Covered	FRS items covered
OQ-01	URS-3.1	FRS-3.1.1,3.1.2,3.2.3
OQ-02	URS-3.2,3.2,3.4	FRS-3.2.1,3.3.1,3.3.2,3.4.1
OQ-03	URS-8.2,8.3	FRS-8.2.1,8.2.2,8.3.1,8.3.2
OQ-04	URS-4.3, URS-2.1	FRS-2.1.1,2.1.2,2.1.3, FRS-4.4.1,4.4.2
OQ-05	URS-4.1	FRS-4.1.1,4.1.2,4.1.3,4.1.4
OQ-06	URS-6.1,6.2,6.3	FRS-6.1.1,6.1.2,6.1.3,6.1.4,6.2.1,6.3.2
OQ-07	URS-6.4	FRS-6.4.1,6.4.2
OQ-08	URS-6.1	FRS-6.1.3
OQ-09	URS-6.3	FRS-6.3.1
OQ-10	URS-2.5	FRS-2.5.1
OQ-11	URS-7.3	FRS-7.3.1,7.3.2
OQ-12	URS-7.1,7.2	FRS-7.1.1,7.1.2,7.2.1
OQ-13	URS-9.1	FRS-9.1.1
OQ-14	URS-2.3,2.4	FRS-2.3.1,2.3.2,2.4.1,2.4.2
OQ-15	URS-2.2	FRS-2.2.1,2.2.2,2.2.3,2.2.4,2.2.5
OQ-16	URS-5.1, URS-8.4	FRS-5.1.1,5.1.2; FRS-8.4.1
OQ-17	URS-8.1; URS-4.2	FRS-8.1.1; FRS-4.2.1,4.2.2,4.2.3

4. Coverage check

Check	Result
URS items in part 1	45
URS items with a FRS item or a stated IQ verification	45 of 45
High-risk items (16) with scripted OQ or PQ test	16 of 16

Medum-risk items (11) with a scripted test or report	11 of 11
Low-risk items (18) with IQ, unscripted check, or review	18 of 18

5. **Known gaps and open items**
    - The IQ and PQ protocols are not yet written, so URS-1.1 to 1.6, URS-9.4 depend on them.
    - The unscripted OQ check OQ checks (URS-3.4, 5.2) need a short record form that lists the check and its outcome.
    - Some FRS items are verified indirectly. Which should get explicit steps when the protocols are next revised.
    - NO test has been executed. This matrix shows planned coverage only.
6. **Approvals**

Role	Name	Signature	date
Prepared by (Validation Engineer)			
Reviewed by (validation Manager)			
Approved by (QA)			